

Tutorial Problem Set X

MATA30 – TUT0003

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In this tutorial we covered L'Hopital's rule and exponential growth and decay. Furthermore, Quiz 4 was written.

1 L'Hopital's rule

Problem 1. Here are some subtle points when applying L'Hopital's rule.

- (a) Why would it be circular in this course to compute $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$ using L'Hopital's rule?
- (b) What is wrong with the following application of L'Hopital?

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x^2} = \lim_{x \rightarrow 0} \frac{\cos(x)}{2x} = \lim_{x \rightarrow 0} \frac{-\sin(x)}{2} = 0$$

Does the limit exist and if so what is its value?

- (c) What is wrong with the following application of L'Hopital?

$$\lim_{x \rightarrow \infty} \frac{x}{x + \sin(x)} = \lim_{x \rightarrow \infty} \frac{1}{1 + \cos(x)} = \text{DNE}$$

Does the limit exist and if so what is its value?

Problem 2. Solve the following problems using L'Hopitals rule.

- (a) Compute $\lim_{x \rightarrow 0^+} x^x$.

- (b) Compute $\lim_{x \rightarrow 0} \frac{x^2}{e^x - x - 1}$.

Problem 3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that

$$\lim_{x \rightarrow \infty} (f(x) + f'(x)) = \lim_{x \rightarrow \infty} e^x f(x) = \infty.$$

Prove that

$$\lim_{x \rightarrow \infty} f'(x) = 0.$$

2 Exponential growth

Problem 4. Let $f(t)$ denote the size of a bacteria population at time t and assume that $f'(t) = 0.05f(t)$ for every $t \in \mathbb{R}$. How long time does it take for the population to grow by a factor of 10?

Problem 5. Let $a, k \in \mathbb{R}$ and consider the initial value problem

$$\begin{cases} y'(x) &= ky(x) \\ y(0) &= a. \end{cases}$$

In this exercise, we are going to find all solutions $y : \mathbb{R} \rightarrow \mathbb{R}$.

- (a) Show that $y(x) = ae^{kx}$ is a solution to the initial value problem.
- (b) Suppose that $y : \mathbb{R} \rightarrow \mathbb{R}$ is a solution. Prove that $(y(x) \cdot e^{-kx})' = 0$ for all $x \in \mathbb{R}$.
- (c) Deduce that $y(x) = ae^{kx}$ for all $x \in \mathbb{R}$. In particular, the initial value problem has a unique solution.

Problem 6. A more realistic model of the growth of a population for many purposes is given by the *Logistic equation*. This takes into consideration that, as the population grows, the competition for resources increases.

$$\begin{cases} y'(x) &= y(x)(1 - y(x)) \\ y(0) &= \frac{1}{2} \end{cases}$$

- (a) Show that $y(x) = \frac{e^x}{e^x + 1}$ is a solution to the initial value problem.
- (b) Suppose that $y : \mathbb{R} \rightarrow \mathbb{R}$ solves the initial value problem. Prove that $y(x) \neq 0$ for all $x \in \mathbb{R}$.
- (c) Prove that $\left(\frac{e^x}{y(x)} - e^x\right)' = 0$ for all $x \in \mathbb{R}$.
- (d) Deduce that $y(x) = \frac{e^x}{e^x + 1}$ for all $x \in \mathbb{R}$. In particular, the initial value problem has a unique solution.